c:1

BASIC NETWORK COMMANDS

AIM:

To execute the basic network commands in windows for network configuration and trouble shooting.

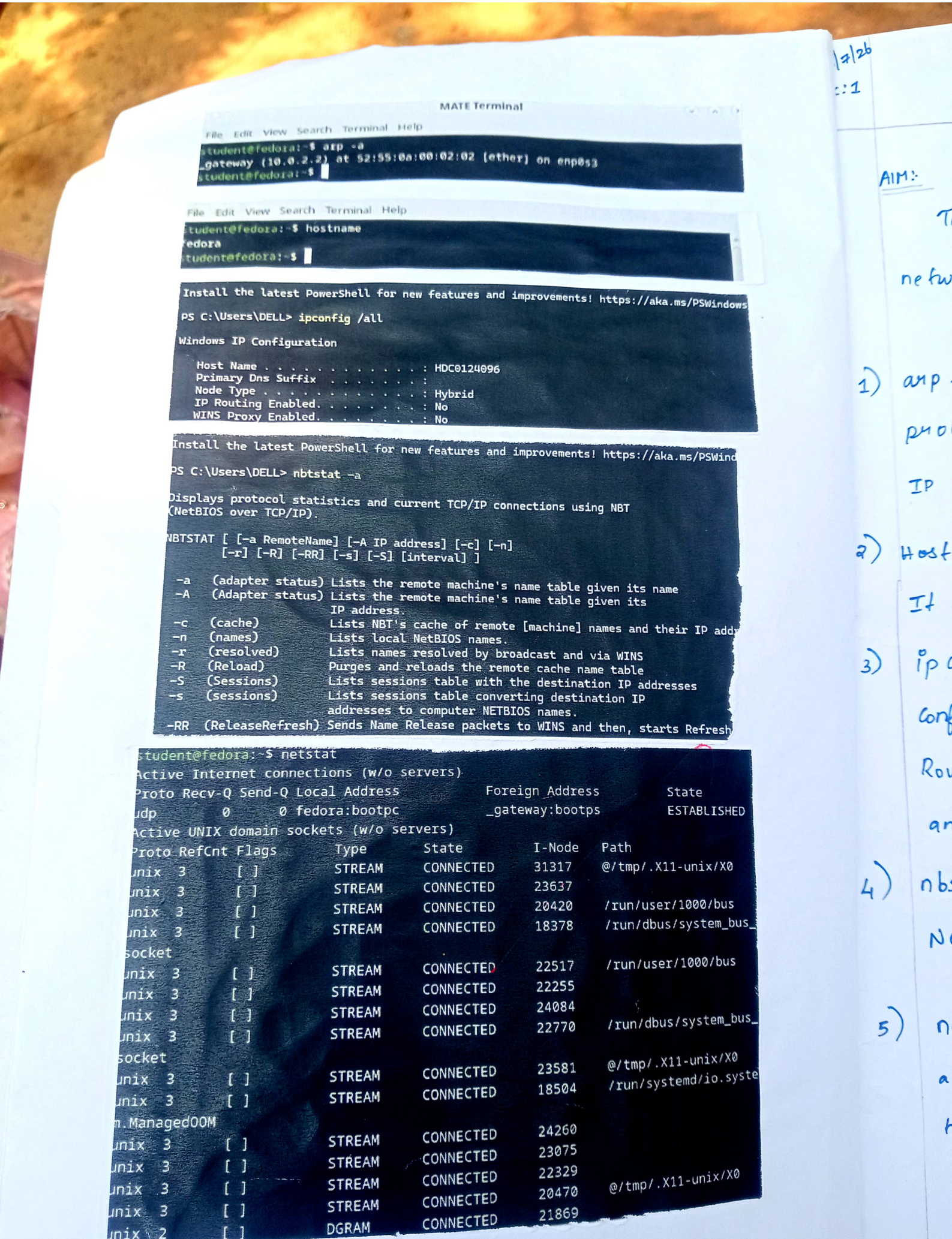
1) arp - a :- ARP is short form of address resolution protocol, It will show the IP computer along with the IP address and MAC address of your router.

2) Hostname :- This is the simplest of all TCP/IP commands It simply displays the computer.

3) ipconfig /all : This command displays detailed configuration information about connection including Routers, Gateway, DNS, DHCP, and type of Ethernet and system.

4) nbstat -a: This command helps solve problems with NetBIOS name resolution for NetBIOS over TCP/IP)

5) netstat - netstat displays a variety of statistics about a TCP/IP connections. It is a command line tool for monitoring network connection incoming and outgoing as well as viewing routing tables, interface statics. eg. netsta -r



MATE Terminal

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```
student@fedora:~$ arp -a
_gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3
student@fedora:~$
```

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```
student@fedora:~$ hostname
fedora
student@fedora:~$
```

```
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\DELL> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : HDC0124096
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
```

```
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWind

PS C:\Users\DELL> nbtstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addr
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh
```

```
student@fedora:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 fedora:bootpc           _gateway:bootps         ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     31317    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     23637
unix  3      [ ]         STREAM     CONNECTED     20420    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     18378    /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     22517    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     22255
unix  3      [ ]         STREAM     CONNECTED     24084
unix  3      [ ]         STREAM     CONNECTED     22770    /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     23581    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     18504    /run/systemd/io.syste
n.ManagedOOM
unix  3      [ ]         STREAM     CONNECTED     24260
unix  3      [ ]         STREAM     CONNECTED     23075
unix  3      [ ]         STREAM     CONNECTED     22329
unix  3      [ ]         STREAM     CONNECTED     20470    @/tmp/.X11-unix/X0
unix  2      [ ]         DGRAM      CONNECTED     21869
```